

Action mechanism of a novel agrichemical quinofumelin against *Fusarium graminearum*

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eLife Assessment

In this **valuable** study, the authors show the physiological response and molecular pathway mediating the effect of quinofumelin, a developed fungicide with an unknown mechanism. The authors present **convincing** data suggesting the involvement of the uridine/uracil biosynthesis pathway, by combining in vivo microbiology characterization as well as in vitro biochemical binding results.

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Abstract

Modern fungicides have made significant contributions to crop disease management, but the development of resistant fungal strains has caused their failure in disease control. Therefore, developing fungicides with novel action mechanisms is the most effective measure to manage resistance. Quinofumelin, a novel quinoline fungicide, exhibits exceptional antifungal activity against phytopathogens. However, there is currently no available information on its mechanism of action. Here, we used transcriptome and metabolome analysis to observe a co-enrichment pattern of differentially expressed genes (DEGs) and differentially accumulated metabolites (DAMs) within pyrimidine biosynthesis pathway (PBP), identifying down-regulation of dihydroorotate dehydrogenase (DHODH). Exogenous UMP, uridine or uracil (metabolites in PBP) successfully restored quinofumelin-induced inhibition of mycelial growth in *Fusarium graminearum* and *Fusarium asiaticum*. Additionally, the deletion of *FgDHODHII* was determined to be lethal; however, mycelial growth of $\Delta FgDHODHII$ mutants could be restored by adding UMP, uridine or uracil. These findings indicate that the deficiencies in *FgDHODHII* are functionally equivalent to complete inhibition of its activity by quinofumelin. Finally, molecular docking, surface plasmon resonance (SPR) and microscale thermophoresis (MST) results strongly support the precise interaction between quinofumelin and *FgDHODHII*. Collectively, these findings provide compelling evidence for the involvement of *de novo* uracil biosynthesis as mechanism of action for quinofumelin while identifying *FgDHODHII* as its specific target.

Introduction

Agrochemical fungicides have a history spanning over 3000 years in the control of plant diseases, and have undergone a development process from traditional protective fungicides with multiple action sites to modern selective fungicides with single action site¹. The introduction of the first fully systemic seed treatment agent (carboxin, a succinate dehydrogenase inhibitor) in 1966 and the launch of the first broad-spectrum foliar systemic fungicide (benomyl, a tubulin polymerization inhibitor) in 1968 represent significant milestones in the advancement of modern selective fungicides with single action site²⁻³. Owing to their exceptional selectivity, potent activity, specific target site, and low toxicity, modern selective fungicides have gained widespread acceptance and witnessed an increasing market share year after year. However, like every coin has two sides, these types of fungicides encounter substantial challenges related to serious resistance issues when deployed in agricultural fields. Numerous reports have been published on plant pathogens developing resistance to modern selective fungicides, which often result in failures or reductions in the effectiveness of chemical control against plant diseases and lead to decreased crop yields and quality (<https://www.frac.info/>)⁴⁻⁶. Therefore, it is crucial to expedite the development of new-generation selective fungicides with innovative modes of action. Based on a single fungicide target, numerous modern selective fungicides can be developed. For instance, 24 commercialized available fungicides have been created so far based on the respiratory chain complex II as the target. Similarly, another 20 commercialized fungicides have been developed targeting the respiratory chain complex III. Additionally, there are already 37 commercialized fungicides that specifically target the fungicide target C14-demethylase in ergosterol biosynthesis (<https://www.frac.info/>). Therefore, discovering novel targets poses a significantly greater challenge compared to creating new fungicides within the existing categories.

According to the latest information released by the Fungicide Resistance Action Committee (FRAC), over 30 fungicide selective targets have been identified and acknowledged in the field of fungicide biological research spanning more than 130 years. Currently, these targets have been identified based on the successful development of fungicides and are all related proteins that are essential for pathogen growth or pathogenicity. Extensive identification and functional analysis of essential genes for growth or pathogenicity in plant pathogens have been achieved with rapid advancements in molecular biology techniques; however, there is no precedent for successfully developing commercialized fungicides targeting genes involved in growth or pathogenicity. The creation of highly active modern selective fungicides currently relies on the structural characteristics of existing targets and is accomplished through technologies such as AI drug discovery and design (AIDD), and computer aided drug design (CADD). The biological functions of drug targets are highly conserved, although variations may occur at critical amino acid sites during genetic evolution. Therefore, it is crucial to explore drug targets and analyze their structural characteristics in order to develop novel fungicides with enhanced activity and selectivity. Recently, the Fungicide Biology group from Nanjing Agricultural University proposed a new concept for targeting pesticides based on the principles of targeted medicine. The objective is to create specialized targeted pesticides that exhibit higher activity, greater selectivity, stronger specialization, lower toxicity, and environmental friendliness by leveraging the structural characteristics of pesticide targets. This approach aims to provide essential technical support for scientifically controlling crop diseases.

Quinofumelin (CAS 861647-84-9) is a novel quinoline fungicide developed by Mitsui Chemicals Co., Ltd. Japan. Due to their significant bioactivity, quinoline compounds and their derivatives have found extensive applications in medicine and pesticides⁷⁻⁹. Based on their biological and physiological characteristics, quinolines are classified as FRAC group 13 (<https://www.frac.info/>). However, the precise mode of action of quinolines against plant pathogens remains unclear. In addition, quinofumelin exhibits no cross-resistance to currently extensively used fungicides,

indicating its unique action target against phytopathogenic fungi. Therefore, this study unveils that the mechanism of action for quinofumelin involves *de novo* synthesis of uracil that identifies dihydroorotate dehydrogenase as its specific target. This newly discovered selective target for antimicrobial agents provides a valuable resource for the design and development of targeted pesticides.

Results

Transcriptomic analysis of *F. graminearum* as affected by quinofumelin

To evaluate the quality of RNA-Seq data, we conducted a comprehensive analysis for quality control. In the principal component analysis, PC1 and PC2 accounted for 45.59% and 16.98% of the total variance, respectively. The distinct separation observed between the groups indicates significant differences in gene expression between the controls and experimental samples, which could be used for subsequent investigations (Supporting Information, Fig. S1). The clustering heat map effectively categorized the controls and experimental samples into distinct clusters based on their gene expression profiles, demonstrating robust biological replication with minimal intra-group variability and ensuring the reliability of the data. Moreover, notable inter-group dissimilarities highlighted variations among different treatment samples (Fig. 1c).

The differential analysis unveiled a total of 234 significantly differentially expressed genes (DEGs) between the control and experimental samples. Among these findings, there were identified as up-regulated for a count of 97 genes while down-regulation was observed in case of another set comprising around 137 genes (Fig. 1a). A graphical representation called “volcano plot” was employed to depict the statistical distribution pattern exhibited by these DEGs (Fig. 1b). To gain deeper insights into gene functionalities, Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analysis were performed on these DEGs. The results showed that they are notably enriched in seventeen distinct GO terms spanning across two major categories; eleven molecular functions along with six processes (Fig. 1d). Furthermore, the enrichment analysis highlighted significant associations with various functional attributes including transition metal ion binding, oxidoreductase activity, modified amino acid binding, amide binding, monooxygenase activity, and heme binding. Moreover, the enriched biological processes primarily involved transcriptional regulation driven by nucleic-acid-enabled mechanisms as well as monocarboxylic acid metabolism and RNA biosynthesis. Additionally, the KEGG enrichment analysis identified specific pathways associated with thiamine metabolism, tryptophan metabolism, nitrogen metabolism, amino acid sugar and nucleotide sugar metabolism, pantothenic acid and CoA biosynthesis, as well as nucleotide sugar production compound synthesis (Fig. 1e).

Metabolomic analysis of *F. graminearum* as affected by quinofumelin

In metabolomics principal component analysis, the contribution rate of PC1 was determined to be 76.6% in positive ion mode and 51.2% in negative ion mode, respectively. Moreover, the quality control (QC) samples exhibited highly concentrated results, indicating excellent instrument stability and remarkable data repeatability. Furthermore, a distinct separation between groups was observed, particularly prominent in the positive ion mode, indicating significant variations in metabolites among different groups that warrant further investigation (Supporting Information, Fig. S2a and S2b). The findings from Pearson correlation analysis were consistent with those obtained from principal component analysis (Supporting Information, Fig. S2c and S2d). Strong correlations were identified within samples belonging to the same group, while weak correlations were observed between different groups.

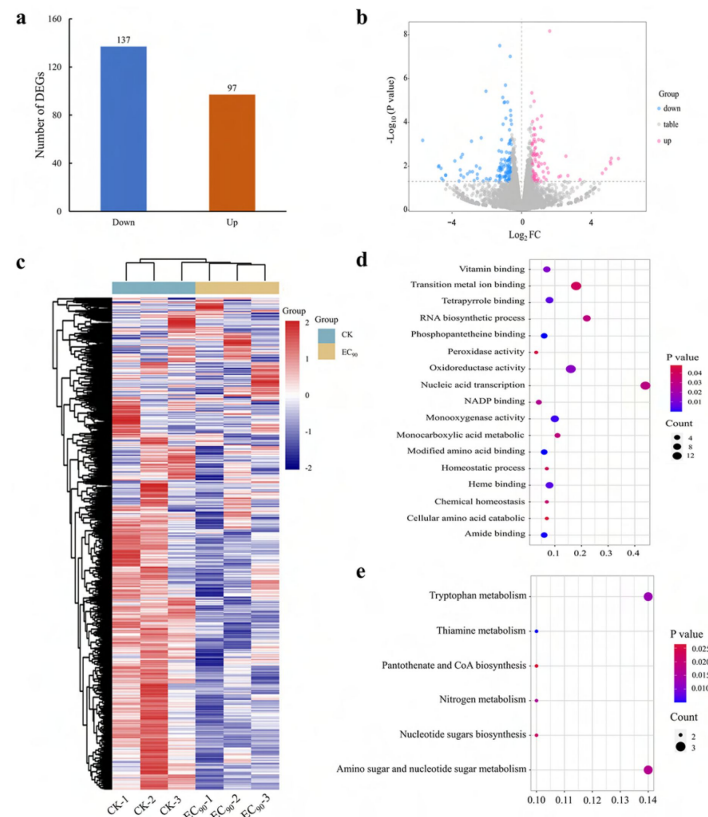


Fig. 1

Analysis of transcriptome sequencing data in *F. graminearum* PH-1 treated with quinofumelin.

- a Bar chart illustrating the comparison of differentially expressed genes between EC⁹⁰ treatments and the control.
- b Volcano plot depicting up-regulated DEGs as red scattered dots and down-regulated DEGs as blue scattered dots.
- c Cluster heat maps displaying gene relative expression values using RNA-seq data scale-standardized values.
- d Bubble map presenting the results of GO enrichment analysis for DEGs.
- e Bubble map showcasing the findings of KEGG enrichment analysis for DEGs.

Metabolomics analysis revealed a total of 713 differential metabolites, with 14 up-regulated and 699 down-regulated in the positive ion mode, as well as 230 differential metabolites, including 103 up-regulated and 127 down-regulated in the negative ion mode (Supporting Information, **Fig. 2a** and Fig. S3a). Based on the classification of the identified 943 different metabolites, it was observed that amino acids and their derivatives accounted for the highest proportion (17%), followed by organic acids and their derivatives (11%), benzene and substituted derivatives (11%), aldehydes, ketones, esters (11%), and heterocyclic compounds (10%) (**Fig. 2c**). The volcano plot visually represents the distribution of statistical differences in metabolite abundance between the two groups (Supporting Information, **Fig. 2b** and Fig. S3b).

KEGG pathways enriched by differential metabolites primarily include arachidonic acid metabolism, pyrimidine metabolism, alanine, aspartate and glutamate metabolism, purine metabolism, citrate cycle, arginine and proline metabolism, fructose and mannose metabolism, nitrogen metabolism, and metabolic pathways (**Fig. 2d**). The classification of enrichment pathways for differential metabolites revealed that the majority of these metabolites were associated with nucleotide metabolism (25%), amino acid metabolism (25%), carbohydrate metabolism (25%), and energy metabolism (12.5%) (Supporting Information, Fig. S3c).

Conjoint analysis of transcriptome and metabolome integration

Through transcriptome and metabolome data analysis, we observed a co-enrichment of differentially expressed genes and differentially accumulated metabolites in the pyrimidine metabolism pathway. In the expression calorigrams of metabolites and genes depicting the *de novo* biosynthesis pathway of pyrimidine, we identified down-regulation of dihydroorotate dehydrogenase (DHODH) in the experimental groups. This down-regulation hinders the conversion of dihydroorotate to orotate, thereby inhibiting the biosynthesis of downstream metabolites orotidine-5P and uracil, leading to their decreased abundance (**Fig. 2e**). Thus, we conclude that the action mechanism of quinofumelin may involve inhibition of the *de novo* biosynthesis pathway of pyrimidine.

Identification of DHODHII in *F. graminearum*

The *FgDHODHII* sequences in *F. graminearum* were identified by performing a BLAST search against the National Center for Biotechnology Information (NCBI) GenBank database, using *Homo sapiens* DHODHII amino acid sequences as a query. *FgDHODHII* is a gene consisting of 1419 bp without introns and is predicted to encode a protein consisting of 472 amino acids. Orthologs of *FgDHODHII* were obtained using the BLAST algorithm. The phylogenetic tree was constructed based on the amino acid sequences of DHODHII homologous protein with Mega X using the neighbor-joining method. Motif pattern information was generated using the MEME suite¹⁰, while functional domain information and protein coordinates were obtained through CD-search¹¹. TBtools software was used for visualizing the tree representation¹². All DHODHII proteins possess the dihydroorotate dehydrogenase domain (**Fig. 3**), indicating high conservation among fungi. Besides, these orthologs proteins would potentially interact with the fungicide in other pathosystems.

Recovery test of mycelial growth suppressed by quinofumelin in *Fusarium* genus

The mycelial growth of *F. graminearum* PH-1 was significantly inhibited on CZA plates containing quinofumelin at concentrations of 1 µg/mL (**Fig. 4**). To investigate the potential recovery of mycelial growth by exogenous dihydroorotate, UMP, uridine, or uracil, CZA plates supplemented with quinofumelin were treated with these compounds. Despite the addition of exogenous dihydroorotate at 50 µg/mL failing to restore mycelial growth, the addition of exogenous UMP, uridine or uracil at 50 µg/mL successfully recovered quinofumelin-induced inhibition of mycelial

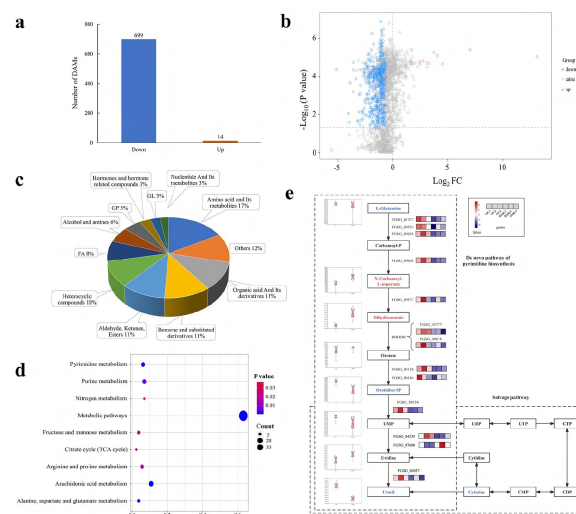


Fig. 2

Difference analysis and enrichment analysis of metabolome data in *F. graminearum* PH-1 treated with quinofumelin.

a Bar chart illustrating the number of differential accumulation metabolites (DAMs) quantity in experimental groups (EC₉₀) and control groups (CK) in positive ion mode.

b Volcanic map of all metabolites in positive ion mode. The red scatter points represent up-regulated DAMs, while the blue scatter points represent down-regulated DAMs.

c Classification and proportion of DAMs.

d Bubble diagram of differential accumulation metabolite KEGG enrichment pathway.

e Pyrimidine *de novo* biosynthesis pathway in *Fusarium graminearum*. The left box plots represent the relative content of 8 metabolite samples, while the right box represents the relative content of 6 gene samples. The color gradient from navy blue to firebrick red indicates a progression from low to high content.



Fig. 3

Phylogenetic tree of DHODHII proteins.

The phylogenetic tree was constructed based on the amino acid sequences of DHODHII homologous protein with Mega X using the neighbor-joining method. The bootstrap values from 1000 replications are indicated on the branches. Motif pattern information was generated using the MEME suite¹², while functional domain information and protein coordinates were obtained through CD-search. The final phylogenetic tree was visualized using TBtools. FgDHODHII was highlighted in red. The amino acid sequence of DHODHII from *Alternaria alternata* (AA0117_g542), *Aspergillus nidulans* (ANIA_05909), *Aspergillus oryzae* (AO1008_05965), *Blumeria graminis* (BLGH_00861), *Botrytis cinerea* (Bcin15g04150), *Botrytis elliptica* (BELL_0162g00150), *Colletotrichum asianum* (GQ607_007053), *Colletotrichum fioriniae* (CFIO01_03831), *Colletotrichum higginsianum* (CH063_10143), *Corynespora cassicola* (BS50DRAFT_576390), *Erysiphe necator* (EV44_g2092), *Erysiphe pulchra* (EPUL_004094), *Fusarium euwallaceae* (BHE90_002825), *Fusarium floridanum* (CEP51_012005), *Fusarium fujikuroi* (FFMR_05040), *Fusarium graminearum* (FGSG_09678), *Fusarium nygamai* (FNYG_10990), *Fusarium odoratissimum* (FOIG_08270), *Fusarium oxysporum* (FOXG_08390), *Fusarium pseudograminearum* (FPSE_04283), *Fusarium solani* (NechaG15121), *Fusarium verticillioides* (FVEG_06288), *Magnaporthe oryzae* (MGG_08814), *Neurospora crassa* (NCU06532), *Oidium neolycopersici* (OnM2_089032), *Pestalotiopsis fici* (PFICI_04984), *Podospira anserina* (PODANS_6_7920), *Puccinia graminis* (PGTG_16739), *Puccinia triticina* (PTTG_02070), *Rhizoctonia solani* (RSOLAG22IIIB_08327), *Schizosaccharomyces pombe* (SPAC57A10.12c), *Ustilagoidea vires* (UVI_02052880), *Verticillium alfalfae* (VDBG_01322), *Verticillium dahliae* (VDAG_07935), and *Zymoseptoria tritici* (Mycgr3G72548) were accessed in EnsemblFungi database.

growth. Similar results were observed in other strains of *F. graminearum* or *F. asiaticum* (Fig. 4 and Fig. S4). Thus, it can be concluded that quinofumelin effectively inhibits the pyrimidine biosynthesis pathways.

The absence of *FgDHODHII* is lethal for mycelial growth in *F. graminearum*

To further investigate the biological functions of *FgDHODHII* in *F. graminearum*, we employed a homologous recombination strategy to generate target gene deletion mutants. Through PCR analysis, two putative *FgDHODHII* deletion mutants were successfully identified. The $\Delta FgDHODHII$ mutants exhibited distinct phenotypic differences compared to the wild-type progenitor PH-1 and the respective complemented strain (Fig. 5a). These mutants displayed an inability to grow on CZA plate; however, their mycelial growth was restored when supplemented with 50 $\mu\text{g/mL}$ of exogenous UMP, uridine or uracil, while no recovery was observed with 50 $\mu\text{g/mL}$ of exogenous dihydroorotate (Fig. 5b).

Molecular docking

To facilitate the molecular docking process between *FgDHODHII* and quinofumelin, ten distinct binding models were generated using AutoDockTools-1.5.7 software, each accompanied by its respective binding energies calculation results. The most favorable conformational arrangement was determined based on its relative ranking in terms of the computed binding energy values and subsequently visualized through PyMOL 2.5.4 software (Fig. 6a). Notably, the obtained docking results revealed a robust interaction between *FgDHODHII* protein and quinofumelin as evidenced by a calculated negative free energy value of -6.8 kcal/mol.

The precise interaction between *FgDHODHII* and quinofumelin was verified by Surface plasmon resonance analysis

Surface plasmon resonance (SPR) was utilized to evaluate the interaction affinity between quinofumelin and *FgDHODHII* in this study. The experiment was conducted using a BIAcore T200 (Cytiva) instrument for biomolecular interaction analysis. *FgDHODHII* was immobilized as the ligand on the chip surface, while quinofumelin continuously flowed through as the analyte solution. Dilute quinofumelin to different concentration gradients (62.5, 32.15, 15.63, 7.81, 3.91 μM), measure the binding energy, and then fit the binding equilibrium dissociation constant (K_d) value (Fig. 6b). The SPR results revealed that quinofumelin exhibited fast-binding and fast-dissociation modes when interacting with the target protein *FgDHODHII*. The calculated K_D value was determined as 6.606×10^{-6} M, indicating specific binding of quinofumelin to *FgDHODHII*.

The interaction between *FgDHODHII* and quinofumelin was confirmed by microscale thermophoresis

In this study, microscale thermophoresis (MST) was further used to evaluate the interaction affinity between *FgDHODHII* and quinofumelin. The strength of the binding can be inferred from the K_d value. The experiment was conducted using MO.Control v1.6.1 and data analysis was performed using MO.Affinity Analysis v2.3. By introducing a fluorescent label GFP to the *FgDHODHII* protein while maintaining its concentration constant, non-fluorescent labeled small molecules were diluted into 16 concentration gradients. Subsequently, the protein solution and small molecule diluent were mixed at a ratio of 19:1 before being loaded into the capillary for further analysis in the machine. The MST assay demonstrated that *FgDHODHII* exhibited specific

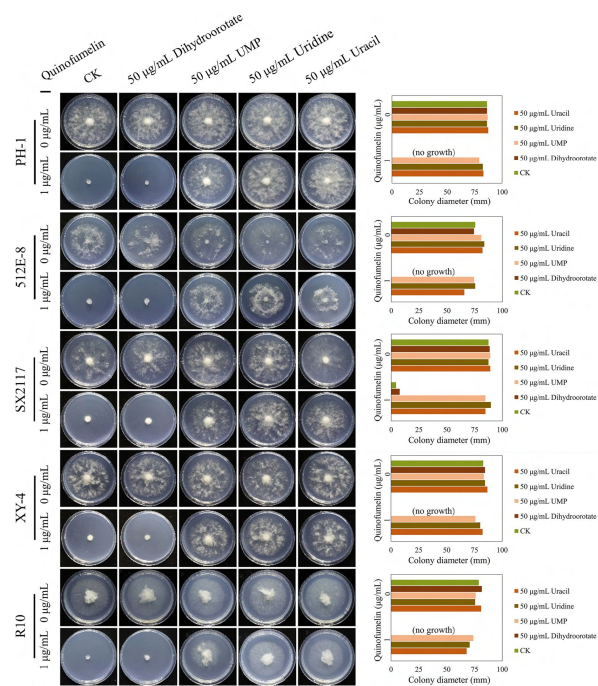


Fig. 4

Recovery test of mycelial growth suppressed by quinofumelin.

All strains were incubated on CZA plates at 25°C for 3 days. *F. graminearum* strains PH-1, 512E-8, SX2117, XY-4, and R10. The left image shows the colony morphology, while the right image is a bar chart of colony diameters.

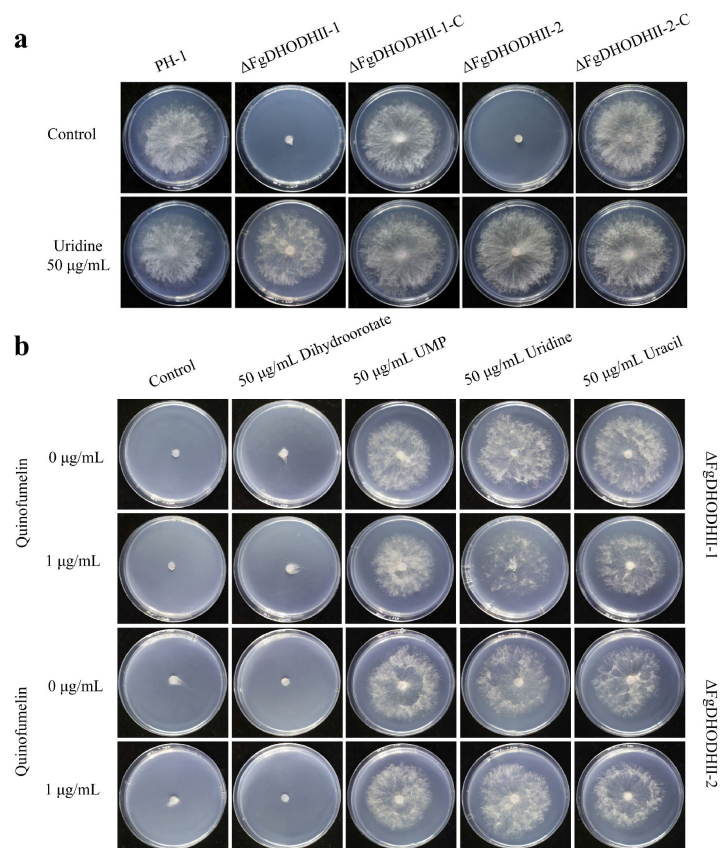


Fig. 5

Mycelial growth of FgDHODHII deletion mutants on CZA plates.

a Mycelial growth of the FgDHODHII deletion mutants and the parental strain was observed in the presence or absence of 50 μ g/mL uridine.

b Recovery of mycelial growth of the FgDHODHII deletion mutants was observed in the presence of 50 μ g/mL dihydroorotate, UMP, uridine or uracil. All strains were incubated at 25°C for 3 days.

binding to quinofumelin in vitro, with a dissociation constant (K_d) of $0.471 \pm 0.268 \mu\text{M}$, whereas no binding was observed with phenamacril which lacks cross-resistance with quinofumelin (**Fig. 6c**).

Discussion

In over 130 years of research on fungicide biology, more than 30 novel targets for selective fungicides have been identified. Based on these newly discovered targets, over 200 modern selective fungicides have been developed. Consequently, the identification of novel targets for fungicides poses a significantly greater challenge compared to the development of new fungicides. Currently, modern selective fungicides with single action site are widely applied in agricultural practices for crop disease management. However, these specific types of fungicides encounter substantial obstacles associated with resistance issues when deployed in agricultural fields. Numerous studies have extensively documented cases of plant pathogens developing resistance to modern selective fungicides, resulting in failures or reductions in their efficacy against plant diseases and subsequently leading to decreased crop yields and quality^{4,6}. For instance, *Botrytis cinerea*, a significant plant pathogenic fungus, has emerged as multi-resistant population that is resistant to all currently used modern selective fungicides. This poses a formidable challenge for the control of crop grey mold disease caused by *B. cinerea*. Therefore, there is an urgent need for the development of new fungicides with novel mechanisms of action to effectively manage multiple resistance.

The novel quinoline, quinofumelin, developed by Mitsui Chemicals Co., Ltd. Japan, has demonstrated remarkable antifungal activity against plant pathogens such as *F. graminearum* and *Sclerotinia sclerotiorum*^{7,13}. Moreover, our research group found that quinofumelin exhibits no cross-resistance to all types of currently used modern selective fungicides. These findings indicate that quinofumelin possesses a distinctive mode of action against plant pathogens. Based on their biological and physiological characteristics, quinolines are classified as FRAC group 13 (<https://www.frac.info/>). However, the precise mechanism by which quinolines act against plant pathogens remains unclear. In this study, we investigated the mechanism of action of quinofumelin on *F. graminearum* using a combined analysis of transcriptomics and metabolomics. The transcriptome data revealed that DEGs were involved in “tryptophan metabolism” and “amino sugar and nucleotide sugar metabolism”. Furthermore, the metabolome data identified DAMs enriched in “nucleotide metabolism” and “amino acid metabolism”. Interestingly, we observed a co-enrichment of DEGs and DAMs in the pyrimidine metabolism pathway. Specifically, we identified a down-regulation of dihydroorotate dehydrogenase (DHODH) in the experimental groups, which impeded the conversion of dihydroorotate to orotate. Consequently, this inhibition led to decreased biosynthesis of downstream metabolites orotidine-5P and uracil. Therefore, we conducted an experiment using *F. graminearum* standard strain PH-1 to quantify mycelial growth by supplementing exogenous uracil, which is the end product of the uracil biosynthesis pathway. This was performed in the presence of complete inhibition of mycelial growth caused by quinofumelin. Surprisingly, the addition of exogenous uracil restored the inhibited mycelial growth induced by quinofumelin (**Fig. 4**). Meanwhile, similar results were successfully verified with four other *F. graminearum* strains and five *F. asiaticum* strains (**Fig. 4** and Fig. S4). Based on our depicted pyrimidine biosynthesis pathway (**Fig. 2E**), we also supplemented other intermediate products such as dihydroorotate, orotate, UMP, and uridine externally. Except for dihydroorotate and orotate, supplementation of exogenous UMP and uridine effectively restored mycelial growth inhibited by quinofumelin as well. These findings strongly support that quinofumelin acts through modulation of the uracil biosynthesis pathway.

As a key enzyme in the *de novo* pyrimidine biosynthesis pathway, DHODH has previously been investigated as a potential target for inhibiting infectious organisms and as an antitumor agent in mammals^{14,15}. Inhibiting DHODH can effectively impede the synthesis of DNA, RNA, and

glycoproteins, thereby regulating aberrant cell metabolism and proliferation to ultimately achieve therapeutic objectives¹⁶. Within mammalian cells, DHODH is closely associated with the mitochondrial membrane and plays a pivotal role in the respiratory chain through its interaction with ubiquinone¹⁷. Here, using amino acid sequences from *Homo sapiens* DHODHII as a reference query, we have identified FgDHODHII (FGSG_09678), a gene comprising of 1419 bp without any introns, which encodes 472 amino acids in *F. graminearum*. FgDHODHII possesses a typical DHOD-2 like domain (NCBI: cd04738) and belongs to class 2 DHODs. Class 2 DHODHs are membrane-bound enzymes, whereas class 1 enzymes are located in the cytosol¹⁸. The DHODH enzyme catalyzes a unique redox reaction in the *de novo* pyrimidine biosynthesis pathway¹⁹. The DHOD-2 like domain can bind orotate and flavin mononucleotide (FMN), facilitating the oxidation of dihydroorotate to orotate and the reduction of the FMN cofactor²⁰. Based on the aforementioned findings, we have concluded that DHODHII functions as the target of action for quinofumelin against *F. graminearum*. Consequently, we performed a protoplast transformation assay to delete FgDHODHII in *F. graminearum* and further validate the precise target of action for quinofumelin. Typically, most fungicide targets are either lethal genes or essential pathogenic genes. Therefore, it is plausible that the inactivation of FgDHODHII may be lethal. In our genetic transformation experiment, exogenous uridine was supplemented to successfully obtain deletion mutants of the FgDHODHII gene. Fortunately, two Δ FgDHODHII mutants were successfully obtained. These mutants exhibited an inability to grow on PDA or CZA medium; however, mycelial growth was fully restored upon addition of uridine to the medium (**Fig. 5A**). Additionally, exogenous supplementation of UMP or uracil effectively restored mycelial growth. However, the supplementation of dihydroorotate to the medium did not alleviate the growth defect observed in Δ FgDHODHII mutants (**Fig. 5B**). Furthermore, mycelial growth in two Δ FgDHODHII mutants was restored by exogenously adding UMP, uracil or uridine under quinofumelin treatments. These findings indicate that the defects of FgDHODHII are functionally equivalent to complete inhibition of its activity by quinofumelin and further confirm that the mechanism of action for quinofumelin involves *de novo* synthesis of uracil while identifying FgDHODH as its specific target.

Molecular docking is a computational procedure employed to predict the optimal binding orientation of a ligand with its macromolecular target (receptor), thereby forming a stable complex. In concise terms, docking is a molecular modeling technique employed to predict the interaction between a protein (enzyme) and small molecules (ligands). In this study, we conducted molecular docking analysis between quinofumelin and the target protein FgDHODHII. The docking results unveiled a robust affinity between FgDHODHII and quinofumelin. Furthermore, these findings provided additional evidence that the strong affinity of quinofumelin with FgDHODHII disrupts its activity and function.

Surface Plasmon Resonance (SPR) technique is employed for real-time measurement of molecular interactions²¹. The SPR sensor map characterizes intermolecular interaction rates primarily by conveying the kinetic rate constant, while it characterizes the binding level between molecules mainly by conveying the affinity constant^{22,23}. In this study, we used SPR technology to measure the binding energy and kinetics between FgDHODHII and quinofumelin. The binding kinetics of FgDHODHII and quinofumelin demonstrated fast-binding and fast-dissociation modes. Soluble FgDHODHII exhibited an ideal affinity (Kd) of 6.606×10^{-6} M with quinofumelin, falling within the suitable range of 10^{-3} M to 10^{-6} M for biomacromolecule-small molecule interactions. Furthermore, the binding affinity between FgDHODHII and quinofumelin was confirmed by a microscale thermophoresis (MST) assay. Therefore, these data strongly support the precise interaction between quinofumelin and its target protein FgDHODHII. This newly discovered selective target for antimicrobial agents offers a valuable resource for the design and development of targeted pesticides. Unlike many other dehydrogenases that require NAD or NADP as an electron acceptor, DHODH transfers electrons to ubiquinone, which then transfers electrons into respiratory complex III and eventually to oxygen²⁴. Thus, the production of orotate in pyrimidine biosynthesis is coupled with the electron transport system and energy production, exerting additional control over nucleotide synthesis and ensuring its coupling to cellular energy

metabolism. In the pyrimidine biosynthesis pathway, DHODHII regulates the conversion from dihydroorotate to orotate. Our findings demonstrated that DHODHII is target of action for quinofumelin. A previous study reported that exogenous orotate but not exogenous dihydroorotate recovered mycelial growth inhibited by quinofumelin, indicating that quinofumelin inhibits the formation of orotate in the pyrimidine biosynthesis pathway²⁵. However, we observed that the supplementation of orotate to the medium did not alleviate the growth defect of Δ FgDHODHII mutants. Considering the disparity in mycelial growth inhibition between *F. graminearum* and other fungi, our data suggests that a nonclassical two-site ping-pong mechanism is the most suitable model for DHODH as previously reported¹⁴. In the classical ping-pong mechanism, DHODH interacts with dihydroorotate and ubiquinol in an identical confirmation, while orotate and ubiquinone combine with an alternative form²⁶. However, similar to brequinar sodium inhibiting the activity of DHODH isolated from mammalian sources, FgDHODHII may follow a nonclassical, two-site ping-pong mechanism. In this nonclassical ping-pong mechanism, the two substrates may not share chemical resemblance and therefore do not bind to a single binding site but likely bind to two adjacent sites that facilitate reactant transfer. The precise nature of the interaction between quinofumelin and FgDHODHII may require further elucidation studies.

Materials and methods

Strains, culture conditions and fungicides

F. graminearum sensitive strain PH-1 and 9 wild-type strains (*F. graminearum* and *F. asiaticum*), and the PH-1 deletion mutants Δ FgDHODHII were maintained in Fungicide Biology Laboratory, Nanjing Agricultural University.

The media used in this study included potato dextrose agar (PDA) medium, Yeast extract peptone dextrose (YEPD) medium, Czapek Solution Agar (CZA) medium and LB medium.

Chemical-grade quinofumelin (Mitsui Agricultural Chemical Company of Japan) was prepared with dimethyl sulfoxide at a concentration of 1×10^4 μ g/mL and stored at 4°C.

Transcriptomic assay of mycelial growth in

F. graminearum as affected by quinofumelin

A total of 20 mL of YEPD medium containing 1 mL of conidia suspension (1×10^5 conidia/mL) was incubated with shaking (175 rpm/min) at 25°C. After 24 h, the medium was added with quinofumelin (EC⁹⁰) at a concentration of 1 μ g/mL, while an equal amount of dimethyl sulfoxide was added as the control (CK). The incubation continued for another 48 h, followed by filtration and collection of hyphae. Subsequently, Genedenovo Biotechnology Co., Ltd. (Guangzhou, China) performed sequencing on the Illumina platform. Quality control measures were implemented to remove linker sequences and low-quality bases from the original reads in order to obtain clean data readings. The filtered data underwent analysis for base composition and mass distribution to visualize data quality. Based on the gene expression data, we used R (<http://www.r-project.org/>) for conducting principal component analysis, while the “pheatmap” package was used to generate the clustering heatmap. DESeq2 was applied to analyze group differences based on quantitative expression results. The screening threshold of P value <0.05 and FC $\geq$ 1.5 or FC $\leq$ 0.67 was used to identify unigenes with significant differential expression between the two groups using a t-test. Finally, functional enrichment analysis of GO and KEGG pathways was performed on these differentially expressed unigenes.

Metabonomic assay of mycelial growth in

F. graminearum as affected by quinofumelin

Transcriptome and metabolome samples were obtained from the same treatment group. The widely targeted metabolome analysis was conducted by Metware Biotechnology Co., Ltd. (Wuhan, China). The frozen sample stored at -80°C was thawed on ice. A 400 μL solution (Methanol: Water = 7:3, v/v) containing an internal standard was added to a 20 mg sample and vortexed for 3 min. Subsequently, the sample was sonicated in an ice bath for 10 min and vortexed for 1 min before being placed at -20°C for 30 min. After centrifugation at 12000 rpm for 10 min (4°C). The sediment was removed, followed by another centrifugation of the supernatant at 12000 rpm for 3 min (4°C). A volume of 200 μL aliquots of supernatant was transferred for LC-MS analysis. All samples were acquired using the LC-MS system according to machine orders. The original data file acquired by LC-MS was converted into mzML format using ProteoWizard software. Peak extraction, peak alignment, and retention time correction were performed by XCMS program.

The identified metabolites were analyzed using MBEole2.0 (<https://csbg.cnbc.csic.es/mbrole2/index.php>) and KEGG (<http://www.kegg.jp>). PLS-DA analysis was used to identify the differentially accumulated metabolites (DAMs), with a screening threshold of P value <0.05 and $\text{FC} \geq 1.5$ or $\text{FC} \leq 0.67$. Each sample was replicated six times for biological consistency.

Recovery assay of quinofumelin-induced

mycelial growth inhibition in *F. graminearum*

F. graminearum sensitive strain PH-1 and 9 wild-type strains (*F. graminearum* and *F. asiaticum*), and the PH-1 deletion mutants $\Delta\text{FgDHODHII}$, were preincubated on a PDA plate to form mycelial colonies. Mycelial plugs (5 mm in diameter) from the edge of a 3-day-old colony were removed and inoculated onto a CZA plate containing quinofumelin (1 $\mu\text{g}/\text{mL}$) and dihydroorotate, UMP, uridine or uracil at 50 $\mu\text{g}/\text{mL}$, followed by measurement of colony diameters. Three replicates were conducted for each concentration.

Construction of deletion mutant vectors

The deletion mutant of *FgDHODHII* was generated through protoplast transformation, as previously described with some modifications in the transformation procedures²⁷. For the transformation process, washed protoplasts were suspended in 160 μL STC and 40 μL SPTC (40% polyethyleneglycol 6000 in STC), along with 2 μg of the aforementioned DNA fragments and 10 μL of heparin (5 mg/mL in STC) to form protoplast suspensions. Subsequently, these mixtures were incubated on ice for 30 min, followed by mixing with 1 mL of SPTC and further incubation at room temperature for 20 min. The resulting suspensions were gently mixed into RM (100 mL) containing 50 $\mu\text{g}/\text{mL}$ uridine and poured into plates. After incubating for 24 h at 25°C , the plates were overlaid with SRM containing 50 $\mu\text{g}/\text{mL}$ uridine and continuously incubated for an additional 4 days. The putative transformants could be observed growing on SRM. All primers used are listed in SI Appendix, Table 1.

Molecular docking

The binding affinity of quinofumelin with *FgDHODHII* protein was assessed through molecular docking using AutoDockTools-1.5.7 software. The three-dimensional structure of quinofumelin (PubChem CID: 23160856) used for docking was obtained from PubChem database, while the protein structure models were sourced from the Uniport database. In the docking studies, binding energy was employed as a metric to evaluate the strength of protein-ligand interaction, where lower values indicated higher affinity between proteins and ligands. The resulting docking outcomes were visualized using PyMOL software version 2.5.4.

Primer	Sequence (5'-3')	Use
FgDHODHII-UF	GGCAGTGAAAGTCTTGTCAAG	PCR primers to amplify FgDHODHII upstream fragment
FgDHODHII-UR	GCTCCTTCAATATCATCTTCTGTGGC	
	TGGGAAGTAATGCTGGG	PCR primers to amplify FgDHODHII downstream fragment
FgDHODHII-DF	GAGACAATACCGAAGGAACAC	
	ATGCTATAGCAAGCAACA	PCR primers to amplify Hph-Hsv-tk fragment
FgDHODHII-DR	AAACTTAACAAATCATCTTTGCC	
SS-F	ACAGAAGATGATATTGAAGGAGC	PCR primers to verify FgDHODHII
SS-R	GAGACAATACCGAAGGAA	
FgDHODHII-YZ-F	TTTGCCCGATCTCTTGGTCT	Constructed FgDHODHII-GFP fusion vector
FgDHODHII-YZ-R	AACCACCAGTCTCCATGAGG	
FgDHODHII-GFP-F	CTCATCACCATCACCATCACAT	PCR primers to construct FgDHODHII expression vector
	GTCTGCCGCTCTCCTACG	
FgDHODHII-GFP-R	AGCTCCTCGCCCTTGCTCACCC	
	TCGAATTCTTCACACCAC	
T7	TAATACGACTCACTATAGG	
T7t	CCGCTAGTTATTGCTCAGC	
pET-28a-F	CCTGGTGCCGCGCGCAGCCATATGCTGG	
	CTGTCAATGTTTTTGAACAG	
pET-28a-R	AAGCTTGTGACGAGCTCGAATTCCTAT	
	TTTAACTCCTGCTTGATCCTG	
pCold-F	GTAAAGCACGCCATATCGC	
pCold-R	CCAAATGGCAGGGATCTTAG	

Table 1

Primers used in the study

Surface plasmon resonance (SPR) verification of specific interactions between FgDHODHII and quinofumelin

The cDNA sequence of FgDHODHII was analyzed and codon optimization was performed for the *E. coli* expression system^{28,29}. Subsequently, it was ligated into the pCold-9×his-MBP plasmid and the pET-28a plasmid at the NdeI and EcoRI sites within the multiple cloning sites to construct recombinant plasmids: pCold-9×his-MBP-TEV-DHODHII³⁰, pET-28a (+)-DHODHII and pET-28a (+)-DHODHII¹²⁴⁻⁴⁶⁷ (with removal of insoluble fragments) (Supporting Information, Fig. S5). Plasmid DNA was extracted using the E.Z.N.A Plasmid Extraction DNA Mini Kit I protocol (Omega, China). For protein overexpression, the exacted plasmids were separately transformed into *E. coli* BL21 strain³¹. A single clone obtained from each transformation was used for protein overexpression and purification. Upon reaching an OD600 of 1.0, 0.6 mM isopropyl β-d-thiogalactopyranoside (IPTG) was added to induce protein overexpression. Following induction at 16°C for 9 h, cells were harvested, lysed, and purified using a Ni-NTA column (QIAGEN).

The interaction between FgDHODHII and quinofumelin was evaluated using Biacore T200 instrument (GE Healthcare). The CM7 sensor chip was employed for immobilizing 50 µg/mL of purified FgDHODHII protein via amino coupling method at a flow rate of 10 µL/min, utilizing 1×PBS-P+ containing 5% DMSO as running buffer. The desired coupling amount of FgDHODHII protein on the chip surface was targeted to be within the range of 12000-24000 RU. To investigate binding affinity, quinofumelin concentrations were diluted from 3.90625 to 62.5 µM in a constant flow rate running buffer containing 5% DMSO. Following sample injection, a switching buffer was introduced to allow spontaneous dissociation of quinofumelin from FgDHODHII on the chip surface, and this dissociation process was continuously monitored in real time by measuring response values. The affinity value (Kd) was determined through kinetic analysis or steady-state affinity method.

Identification of specific interactions between FgDHODHII and quinofumelin using microscale thermophoresis (MST)

To construct the FgDHODHII-GFP cassette, the FgDHODHII fragment containing the native promoter and ORF (without the stop codon) was amplified. Subsequently, the resulting PCR product was co-transformed with XhoI-digested pYF11 into *E. coli* DH5α as previously reported³². The fusion vector was then used to transform PH-1 to obtain the FgDHODHII-GFP strain. The GFP fusion protein was extracted from mycelia. Two Monolith NT.115 standard capillaries were employed for repeated absorption of a specific protein concentration, and placed in positions 1 and 2 of the sample tray in the MonoTemper Monolith NT.115 instrument. Sixteen PCR tubes were prepared, with 10 µL DMSO added to PCR tubes 2-16 and 20 µL of 5000 µM quinofumelin (or phenamacril) dissolved in DMSO added to PCR tube 1. Then, 10 µL of quinofumelin from PCR tube 1 was transferred and thoroughly mixed with PCR tube 2 using a pipette. These steps were repeated for PCR tubes 3–16. Following mixing with PCR tube 16 (discarding the remaining 10 µL), each PCR tube was thoroughly mixed with 190 µL DHODHII-GFP protein, and incubated at room temperature for 30 min. Monolith NT.115 standard capillaries were inserted into each PCR tube to allow solution entry into the capillary. The capillaries 1-16 were subsequently placed in the sample tray of the device and the program was initiated. All measurements were performed using 80% (Auto-detect) Nano-RED. MonoTemper Monolith NT.115 software was used to evaluate the binding affinity between the target protein and quinofumelin using the Kd values and the signal to noise ratio, and the results were combined and analyzed using Mo. Affinity Analysis v2.3 software³³.

Statistical analysis

The data in the study were analyzed using analysis of variance (ANOVA) with the SPSS 14.0 software (SPSS Inc. Chicago, IL, USA). When ANOVA yielded a significant result ($p = 0.05$), means were compared using Fisher's protected least significant difference (PLSD).

Data availability

Data will be made available on request.

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Reviewer #1 (Public review):

Summary:

Phytophathogens including fungal pathogens such as *F. graminearum* remain a major threat to agriculture and food security. Several agriculturally relevant fungicides including the potent Quinofumelin have been discovered to date, yet the mechanisms of their action and specific targets within the cell remain unclear. This paper sets out to contribute to addressing these outstanding questions.

Strengths:

The paper is generally well-written and provides convincing data to support their claims for the impact of Quinofumelin on fungal growth, the target of the drug, and the potential mechanism. Critically the authors identify an important pyrimidine pathway dihydroorotate dehydrogenase (DHODH) gene *FgDHODHII* in the pathway or mechanism of the drug from the prominent plant pathogen *F. graminearum*, confirming it as the target for Quinofumelin. The evidence is supported by transcriptomic, metabolomic as well as MST, SPR, molecular docking/structural biology analyses.

Weaknesses:

Whilst the study adds to our knowledge about this drug, it is, however, worth stating that previous reports (although in different organisms) by Higashimura et al., 2022 <https://pmc.ncbi.nlm.nih.gov/articles/PMC9716045/> had already identified DHODH as the target for Quinofumelin and hence this knowledge is not new and hence the authors may want to tone

down the claim that they discovered this mechanism and also give sufficient credit to the previous authors work at the start of the write-up in the introduction section rather than in passing as they did with reference 25? other specific recommendations to improve the text are provided in the recommendations for authors section below.

<https://doi.org/10.7554/eLife.105892.1.sa2>

Reviewer #2 (Public review):

Summary:

In the current study, the authors aim to identify the mode of action/molecular mechanism of characterized a fungicide, quinofumelin, and its biological impact on transcriptomics and metabolomics in *Fusarium graminearum* and other *Fusarium* species. Two sets of data were generated between quinofumelin and no treatment group, and differentially abundant transcripts and metabolites were identified. The authors further focused on uridine/uracil biosynthesis pathway, considering the significant up- and down-regulation observed in final metabolites and some of the genes in the pathways. Using a deletion mutant of one of the genes and in vitro biochemical assays, the authors concluded that quinofumelin binds to the dihydroorotate dehydrogenase.

Strengths:

Omics datasets were leveraged to understand the physiological impact of quinofumelin, showing the intracellular impact of the fungicide. The characterization of FgDHODHII deletion strains with supplemented metabolites clearly showed the impact of the enzyme on fungal growth.

Weaknesses:

Some interpretation of results is not accurate and some experiments lack controls. The comparison between quinofumelin-treated deletion strains, in the presence of different metabolites didn't suggest the fungicide is FgDHODHII specific. A wild type is required in this experiment.

Potential Impact: Confirming the target of quinofumelin may help understand its resistance mechanism, and further development of other inhibitory molecules against the target.

The manuscript would benefit more in explaining the study rationale if more background on previous characterization of this fungicide on *Fusarium* is given.

<https://doi.org/10.7554/eLife.105892.1.sa1>

Reviewer #3 (Public review):

Summary:

The manuscript shows the mechanism of action of quinofumelin, a novel fungicide, against the fungus *Fusarium graminearum*. Through omics analysis, phenotypic analysis, and in silico approaches, the role of quinofumelin in targeting DHODH is uncovered.

Strengths:

The phenotypic analysis and mutant generation are nice data and add to the role of metabolites in bypassing pyrimidine biosynthesis.

Weaknesses:

The role of DHODH in this class of fungicides has been known and this data does not add any further significance to the field. The work of Higashimura et al is not appreciated well enough as they already showed the role of quinofumelin upon DHODH II.

There is no mention of the other fungicide within this class ipflufenquin, as there is ample data on this molecule.

<https://doi.org/10.7554/eLife.105892.1.sa0>